

f) Ionic compounds

*Students will be assessed on their ability to:*

- 1.27 describe the formation of ions by the gain or loss of electrons
- 1.28 understand oxidation as the loss of electrons and reduction as the gain of electrons
- 1.29 recall the charges of common ions in this specification
- 1.30 deduce the charge of an ion from the electronic configuration of the atom from which the ion is formed
- 1.31 explain, using dot and cross diagrams, the formation of ionic compounds by electron transfer, limited to combinations of elements from Groups 1, 2, 3, and 5, 6, 7
- 1.32 understand ionic bonding as a strong electrostatic attraction between oppositely charged ions
- 1.33 understand that ionic compounds have high melting and boiling points because of strong electrostatic forces between oppositely charged ions